Smart Horizon 2026:
48-Hour International Hackathon

Grand Finale

Organized by: Dept. of Artificial Intelligence and Machine Learning & Dept. of Computer Science and Engineering

Event Date	Venue	Host Institution
September 3rd, 4th & 5th, 2026	New Horizon Knowledge Park, Bangalore, Karnataka, India	New Horizon College of Engineering

TITLE: Automated text reading system by OCR and text to speech

Team Members

1.DEEKSHITA TAMARAPALLI
2.ABHINAV KUMAR
3.HIMESH KUMAR CHANDORA
4.CHIRAG PRAKASH OSWAL
5.CHANDANA P V

Team Details

Team ID: SHIH26-TID-540
Team Name: DBC
Team Lead: DEEKSHITA TAMARAPALLI
College Name: NEW HORIZON COLLEGE OF ENGINEERING
Problem ID: SH-SVA-05

The Problem Understanding

Challenges faced by visually impaired individuals in reading printed text

2.2B+ People worldwide live with vision impairment	**90%** Of visually impaired live in low-income countries	**80%** Of visual impairments are preventable or treatable

Key Challenges

- **Cannot read books, labels, signs, or documents independently**
- **Difficulty reading medicine labels and dosage instructions — a safety risk**
- **Limited access to printed educational material in schools and colleges**
- **Workplace challenges — unable to read printed reports or notices**
- **Existing solutions are expensive, bulky, or require internet access**

Our Target Users / Stakeholders

People with visual impairments

Blind individuals and those with low vision who cannot read printed text independently. This is the core audience the solution is built for.

Elderly people

Elderly people with deteriorating eyesight who struggle with small print like medicine labels, bills, or newspapers.

Students with visual disabilities

Students with visual disabilities who need to read textbooks, exam papers, and study materials without depending on others.

Hospital patients

Hospital patients who need to read prescriptions, dosage instructions, or medical reports on their own.

Workplace professionals

Workplace professionals with vision impairment who deal with printed documents, reports, or office materials daily.

Literature Survey

Existing Solution	What it does	Limitation / Gap
Seeing AI	Reads text, recognizes objects/scenes and provides audio feedback	Primarily smartphone-based; not designed around dedicated smart-glasses architecture
Be My Eyes	Connects visually impaired users with sighted volunteers	Requires human assistance and internet connectivity
Google Lookout	Identifies text, objects, images and documents	General-purpose assistance; limited personalized storytelling experience
Envision	OCR, object recognition and visual assistance	Advanced features may require dedicated hardware/subscription
Research on Assistive Wearables	Uses cameras + AI for environmental awareness	Often focuses on individual functions rather than an integrated, low-power system

PROPOSED SOLUTION

VisionRead is an AI-powered assistive vision system designed to help visually impaired users understand text and their surroundings through hands-free audio feedback.

How it works

Camera → AI Processing → Understanding → Prioritization → Voice Feedback

- **Camera:** Captures books, documents, signs and the surrounding environment.
- **AI Processing:** Uses OCR and computer vision to understand what the camera sees.
- **Text Recognition:** Extracts readable text from physical books, documents and messages.
- **Environmental Awareness:** Identifies relevant road signs, hazards and objects.
- **Priority-Based Alerts:** Important hazards/signs can be communicated before less important information.
- **Voice Feedback:** Converts information into audio so the user doesn't need to look at a screen.
- **Story Mode:** Converts longer text into more natural, expressive audiobook-style narration using Gemini TTS.
- **Voice Control:** Allows the user to control scanning, reading, playback and modes through voice commands.

Novelty of Solution

VisionRead introduces a unified, hands-free assistive system that combines visual understanding with intelligent audio interaction.

1. Glasses-as-Eyes, Phone-as-Brain Architecture

- Smart glasses are designed primarily for **visual capture**.
- The smartphone performs the computationally intensive AI processing.
- This approach can make future wearable hardware **lighter, simpler and more power-efficient**.

2. Two-Level Audio Experience

- **Normal Mode:** Fast Android TTS for everyday text and visual information.
- **Story Mode:** Gemini TTS provides more expressive, audiobook-style narration for books and long-form content.
- This moves beyond simply *reading text* toward **making long-form content more engaging and accessible**.

3. Context-Aware Prioritization

Instead of announcing every detected object equally, VisionRead is designed to prioritize information that matters most to the user, particularly **hazards and road-related information**.

4. Voice-First, Screen-Independent Interaction

Users can control core functions through voice commands such as **Scan, Repeat, Stop, Faster, Slower, Pause, Resume, Story Mode and Normal Mode**, reducing dependence on visual interfaces.

5. One Platform, Multiple Assistive Functions

VisionRead brings together:

OCR + Environmental Awareness + Priority Alerts + Voice Control + Expressive Reading

Strong one-line novelty statement

Dataset Used

Function	Technology / Model	Dataset
Text Recognition	Google ML Kit Text Recognition	Pre-trained ML Kit model
Object Detection	Google ML Kit Object Detection	Pre-trained ML Kit model
Road Sign Detection	VisionRead RoadSignDetector	Implemented using the existing vision-processing pipeline; no custom training dataset in the current prototype
Hazard Detection	VisionRead HazardDetector	Implemented using the existing vision-processing pipeline; no custom training dataset in the current prototype
Story Narration	Gemini TTS	Google's pre-trained generative audio model
Speech Commands	Android SpeechRecognizer	Platform speech-recognition service
Normal Voice Output	Android Text-to-Speech	Platform TTS engine

Technology Stack

Development
- **Java**
- **Android Studio**
- **Gradle / Android SDK**

Computer Vision
- **CameraX** — camera capture
- **Google ML Kit Text Recognition** — OCR
- **Google ML Kit Object Detection** — object/environment detection

Audio & AI
- **Android SpeechRecognizer** — voice commands
- **Android Text-to-Speech** — Normal Mode
- **Gemini TTS** — Story Mode / expressive narration

Communication
- **Wi-Fi / TCP**
- **Bluetooth**

Architecture
- **Smart Glasses → Camera / Input**
- **Smartphone → AI Processing**
- **Audio → User**

Technical Architecture

SMART GLASSES / CAMERA
|
| Wi-Fi / Bluetooth
▼
SMARTPHONE — VISIONREAD BRAIN
|
▼
VISION PROCESSING
CameraX • ML Kit OCR
Object • Sign • Hazard
Detection
|
▼
DECISION & ALERT LAYER
Context Processing
Priority-Based Alerts
|
▼
AUDIO ENGINE
Normal → Android TTS
Story → Gemini TTS
|
▼
AUDIO OUTPUT
Phone / Glasses
|
▼
USER

Technical Architecture

Main Components

1. Input Layer

Camera captures the user's visual environment.
Physical controls and voice commands provide user input.
Camera input can originate from the smartphone or the future smart-glasses setup.

2. Communication Layer

Wi-Fi/TCP
Bluetooth
Handles communication between the smartphone and glasses.

3. Vision Processing Layer

CameraX manages camera frames.
ML Kit Text Recognition extracts text.
ML Kit Object Detection identifies objects.
RoadSignDetector handles road-sign recognition.
HazardDetector handles hazard-related detection.

4. Decision & Alert Layer

AlertManager manages important detected information.
Information is prioritized so critical alerts can be communicated appropriately.

5. Audio Intelligence Layer

Normal Mode
Detected text → Android TTS → Audio
Story Mode
Detected book text → Gemini TTS → PCM audio → Audio playback

6. Output Layer

Audio is delivered to the user through the phone's audio output.
The existing glasses communication architecture can also communicate information between devices.

Implementation

1. <u>Android Application</u>

Developed using **Java in Android Studio**.
Modular architecture separates camera input, vision processing, communication, alerts and audio output.
The smartphone acts as the primary **processing/brain unit**.

2. <u>Camera & Vision Processing</u>

CameraX captures camera frames.
Frames are processed using **Google ML Kit**.
Text Recognition (OCR) extracts text from books, documents, signs and messages.
Object Detection identifies relevant objects in the environment.
Road-sign and hazard detection modules process environmental information.

3. <u>Intelligent Reading</u>

Detected text is checked for stability before triggering reading, helping avoid repeatedly reading incomplete camera frames.
In **Normal Mode**, recognized text is spoken using **Android Text-to-Speech**.
In **Story Mode**, recognized long-form text is sent to **Gemini TTS** for expressive audiobook-style narration.

4. <u>Environmental Alerts</u>

Detected signs and hazards are passed to the **Alert Manager**.
Alerts are prioritized so important information can receive appropriate attention.
The system provides audio feedback instead of requiring the user to look at the phone.

5. <u>Voice Control</u>

The application uses **Android SpeechRecognizer** to support commands such as:
Scan → Repeat → Stop → Faster → Slower → Pause → Resume → Story Mode → Normal Mode

6. <u>Smart Glasses Communication</u>

The architecture separates the **camera/input device** from the **processing device**.
Wi-Fi/TCP and Bluetooth communication are used for device connectivity.
This supports the future concept of lightweight glasses acting primarily as the user's **eyes**, while the smartphone performs the computation.

Prototype

(App icon)

VisionRead

OCR Assistant for the Visually Impaired

An Android application that uses AI-powered text recognition and voice guidance to help visually impaired individuals read printed text hands-free.

1. Live Camera OCR
2. Text-to-Speech
3. Voice Guidance

Feasibility

1. Technical Feasibility

Built on existing Android smartphone hardware and software.

Uses established technologies such as **CameraX, Google ML Kit, Android SpeechRecognizer and Android TTS**.

Gemini TTS can provide expressive narration for Story Mode.

Wi-Fi/Bluetooth enables communication between the phone and the proposed glasses.

Modular architecture allows individual components to be improved independently.

2. Economic Feasibility

Prototype can be developed using commercially available smartphones.

No specialized AI-processing hardware is required for the current prototype.

Future glasses can be designed mainly for **camera capture and communication**, reducing onboard computational requirements.

Cloud-based AI services introduce API costs, so production deployment would require cost optimization.

3. Operational Feasibility

Designed around **hands-free, voice-first interaction**.

Users can control reading and narration using simple commands.

Audio feedback eliminates the need to continuously interact with a screen.

Normal Mode provides quick reading, while Story Mode is designed for longer content.

4. Scalability & Future Feasibility

The architecture can be extended with:

Lightweight dedicated smart glasses

Better road-sign and hazard models

Distance and direction estimation

Scene understanding

Multilingual support

Offline AI/TTS ,More efficient, low-power processing

Impact

1. Greater Independence

Helps visually impaired users access **books, documents, labels and messages** without depending on another person.

Enables hands-free access to visual information.

2. Better Access to Education

Makes physical books and printed learning material more accessible.

Story Mode can make long-form reading more engaging through audiobook-style narration.

Supports a more natural listening experience than conventional monotone reading.

3. Improved Environmental Awareness

Provides audio information about **road signs, objects and potential hazards**.

Priority-based alerts can help users focus on more important information.

4. Accessible Human–Computer Interaction

Voice commands reduce dependence on visual smartphone interfaces.

Users can control scanning and audio playback through simple commands.

5. Reduced Cognitive Load

Instead of continuously processing everything visually, the system aims to:

Detect → Understand → Prioritize → Communicate

This can help deliver only the information that is relevant at that moment.

6. Long-Term Social Impact

VisionRead aims to contribute toward:

Greater digital and physical accessibility

More independent mobility

Improved access to information

More inclusive education

More affordable assistive technology

Results

1. Text Recognition & Reading

Successfully captures physical text through the camera.

OCR extracts readable text using **Google ML Kit**.

Detected text can be converted into speech automatically.

Stable-frame detection helps prevent reading incomplete camera frames.

2. Dual Reading Modes

Normal Mode: Provides fast, functional reading using Android TTS.

Story Mode: Converts book/long-form text into more expressive audiobook-style narration using Gemini TTS.

Supports **pause, resume, repeat, faster and slower** playback.

3. Environmental Awareness

Prototype includes processing for **road signs and hazards/objects**.

Important information can be handled through the alert-management layer.

Audio feedback allows information to be delivered without relying on the screen.

4. Voice-First Control

Core functions can be controlled through voice commands.

Users can switch between reading modes and control playback hands-free.

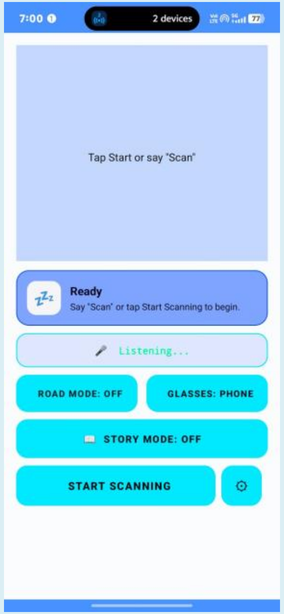
5. Eyes–Brain Architecture Demonstrated

Smartphone can act as the **AI processing unit**.

Communication architecture supports the concept of separating the camera/inputs from the processing device.

Screenshots

Home Screen

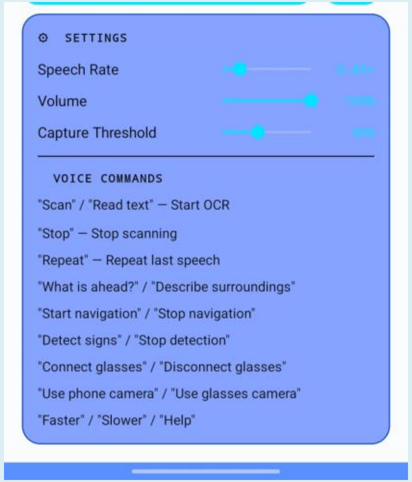


Normal mode reading

Story mode reading

Screenshots

Settings panel

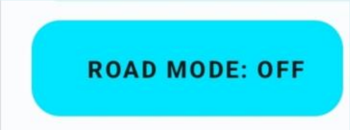


Device pairing panel

Eye mode interface

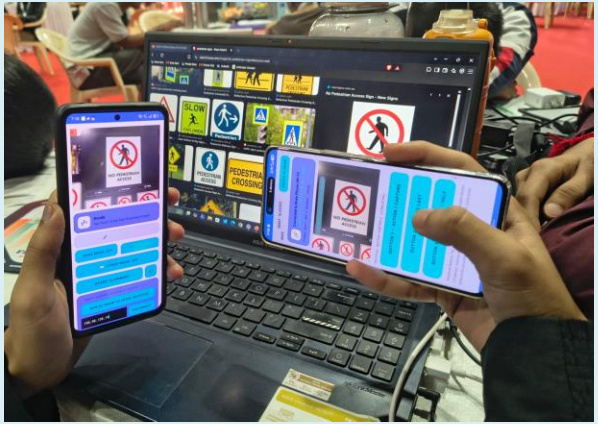
Screenshots

Road mode



Story mode

Two device working together



Future Enhancements

1. Dedicated Lightweight Smart Glasses

Move from smartphone-camera testing to custom glasses with a small camera.
Glasses act primarily as the **"eyes"**, while the smartphone remains the **"brain."**
Reduce weight, power consumption and hardware complexity.

2. Real-Time Distance & Direction Detection

Estimate how far a detected obstacle is from the user.
Provide directional alerts such as:

- "Obstacle ahead"
- "Vehicle approaching from the left"
- "Clear path on the right"

3. Advanced Scene Understanding

Move beyond detecting individual objects.
Understand the **overall scene and context**.
Example: distinguish between a pedestrian crossing, road intersection, entrance, stairs, etc.

4. Improved Road & Hazard Detection

Develop and train custom datasets for:

- Indian road signs
- Vehicles
- Pedestrians
- Potholes
- Construction barriers
- Open drains and other hazards

Improve detection accuracy under different lighting and weather conditions.

5. Automatic Page/Book Progression

Maintain reading position across multiple pages.
Detect when a new page is presented.
Automatically continue narration instead of requiring the user to repeatedly trigger scanning.

6. More Advanced AI Narration

Support multiple languages and regional accents.
More natural character/dialogue differentiation.
Context-aware narration for longer books.
Better handling of very long documents through intelligent text chunking.

7. Offline AI & TTS

Reduce dependence on internet connectivity.
Run lightweight OCR, detection and speech models locally where possible.
Improve privacy, latency and reliability.

8. Direct Audio Streaming to Glasses

Extend the current communication architecture to transmit generated audio directly to the glasses.
Enable the glasses to provide the final audio output without requiring the phone speaker.

Conclusion

VisionRead is an AI-powered assistive vision system designed to transform visual information into accessible audio for visually impaired users.

Key Takeaways

- **<u>Understand the Visual World</u>**
  Uses computer vision and OCR to interpret text, objects, road signs and hazards.
- **<u>Make Information Accessible</u>**
  Converts detected information into voice feedback, reducing dependence on visual interfaces.
- **<u>Go Beyond Robotic Reading</u>**
  Story Mode uses Gemini TTS to provide more natural, expressive audiobook-style narration for books and long-form text.
- **<u>Scalable Eyes–Brain Architecture</u>**
  The smartphone performs the computational work while future lightweight glasses can serve primarily as the camera/input device.
- **<u>Designed for Independence</u>**
  The overall goal is to help users access information and understand their surroundings more independently through a hands-free, voice-first experience.

THANK YOU